Dmitrii **Altukhov**

Senior Python AI Engineer · PhD (Computer Science)

Amsterdam, The Netherlands

dmalt | dmitrii-altukhov-258b05233 | altukhov.dm@gmail.com | +316 136 36 770

Profile

Senior Python Engineer with 13+ years of software engineering experience building production AI systems and scalable backend services. PhD in Computer Science. Track record of designing event-driven architectures, deploying generative AI pipelines, and owning AI features end-to-end from prototyping through production.

Python	FastAPI, AsyncIO, Pydantic, Poetry, pytest
AI & ML	LLM integration, AI agents, RAG pipelines, prompt engineering, evaluation, PyTorch
Infrastructure	AWS, Kubernetes, Docker, Kafka, Redis, PostgreSQL, CI/CD, Git

Experience

Creative Fabrica — *Amsterdam, the Netherlands*

Python Engineer → ML Engineer → Senior ML Engineer — *Dec. 2023 – present*

- Own the generative AI **Python** codebase, infrastructure, and internal testing and QA tool.
- Developed an AI agent for canvas manipulation and built an evaluation pipeline to benchmark against the baseline agent.
- Developed and deployed to production an inference pipeline for content matching.
- Designed and maintained distributed event-driven architecture using **Kafka**, **Redis** queues, and **Kubernetes**.
- Redesigned the architecture for image generation backend on **AWS**, significantly improving scalability and system performance.

SBDA Group — *Remote*

Data Scientist (contract) — *Sep. 2023 – Oct 2023*

- Designed and implemented an algorithmic solution in Python for optimizing a manufacturing supply chain process.

Artificial Intelligence Research Institute (AIRI) — *Moscow, Russia*

Research fellow — *Feb. 2022 – Apr. 2023*

- Built and trained a deep learning model to predict speech from brain data in **PyTorch**.
- Developed real-time **Python** pipeline for EEG timeseries prediction using Kalman filtering.
- Developed a **Python** library for type-safe timeseries arrays manipulation.

Higher School of Economics, Centre for Cognition and Decision Making — *Moscow, Russia*

Junior research fellow — *Feb. 2015 – Dec. 2021*

- Accelerated various parts of my PhD codebase to enable Monte-Carlo statistics, using **CUDA** (cuBLAS) and **C**.
- Led 4-people group on a project for real-time ML feature extraction and GPU-accelerated 3D visualization of brain activity in **Python**.
- Sped up by a factor of 10 the beamformer inverse solver in MNE-python by modifying the algorithm for vectorized computations.

University of Montreal, CERNEC lab. — *Montreal, Canada*

Visiting researcher, temporary position — *Oct. – Dec. 2015, May 2016 – Dec. 2016*

- Co-developed an open-source **Python** package for GPU-accelerated brain data visualization Visbrain, Combrisson et al. [2019].

Education

Ph.D. in Computer Science — *Moscow, Russia*

Higher School of Economics, Faculty of Computer Science — *Jan. 2016 – Nov. 2021*

GPA: 4.0; proposed an algorithmic framework for brain connectivity estimation; research published in the leading domain journals.

Specialist degree in Mechanics *(Masters equivalent)* — *Moscow, Russia*

Lomonosov Moscow State University, Department of Mechanics and Mathematics — *Sep. 2008 – Jun. 2013*

Thesis: “Enhancement and validation of LOGOS software for simulations of the reactive fluid flows.”

Algorithms, Part 1 and Part 2 — *Online*

Coursera, Princeton University — *2020*

Comprehensive Algorithms and Data Structures course in **Java**.